Enhanced ROS for humanoid robots (Nasa Robonaut 2) constructing habitats

OPERATING SYSTEMS PROJECT REPORT

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GitHub link: https://github.com/vidhya041007?tab=repositories

https://github.com/aakankshakand30-debug?tab=repositories

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BONAFIDE CERTIFICATE

Certified that the project titled "ENHANCED ROS FOR HUMANOID ROBOTS (NASA ROBONAUT 2) CONSTRUCTING HABITATS" is the project work of AAKANKSHA K(RA2511026020146),H VIDHYA LAKSHMI (RA2511026020159), of II Year CSE-AIML, prepared for the Operating Systems course for the Academic Year 2026–27.

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ABSTRACT

The integration of an enhanced Robot Operating System (ROS) with humanoid robots, particularly NASA's Robonaut 2 (R2), offers a promising approach to the autonomous construction of habitats in space. As space exploration advances towards long-duration missions on the Moon and Mars, the need for intelligent robotic systems capable of performing complex construction tasks in hazardous and resource-constrained environments is increasing. ROS provides a flexible, modular, and open-source framework that enables communication between robotic sensors, actuators, perception systems, motion-planning algorithms, and artificial intelligence modules. An enhanced ROS architecture can improve the capabilities of humanoid robots by incorporating advanced computer vision, real-time sensor fusion, autonomous navigation, precise manipulation, obstacle detection, and adaptive decision-making. Using these technologies, robots inspired by Robonaut 2 can identify construction materials, handle tools, assemble structural components, install habitat modules, and assist in the deployment of essential infrastructure with minimal human intervention. Integration with digital simulations and digital twin environments can further support mission planning, virtual testing, and the prediction of potential construction failures before physical operations begin. Moreover, enhanced communication protocols and fault-tolerant control mechanisms can improve operational reliability when communication delays, limited connectivity, extreme temperatures, radiation, and reduced gravity present challenges. Human–robot collaboration can also allow astronauts to supervise robotic activities, assign construction tasks, and intervene when complex decisions are required. Although Robonaut 2 was primarily developed to demonstrate advanced humanoid robotic capabilities in space rather than as a dedicated autonomous habitat-construction system, its dexterity, adaptability, and experience in human-oriented tasks provide a valuable foundation for future robotic construction platforms. Therefore, the proposed enhanced ROS-based framework aims to develop safer, more efficient, and more autonomous humanoid robotic systems capable of supporting the construction, maintenance, and expansion of extraterrestrial habitats, ultimately reducing astronaut risk, minimizing operational costs, and enabling sustainable human settlement beyond Earth.

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1. INTRODUCTION

Space exploration and extraterrestrial colonization require robust robotic systems capable of operating autonomously or semi-autonomously in hazardous environments. NASA's **Robonaut 2 (R2)**—a highly dexterous humanoid robot originally deployed on the International Space Station (ISS)—serves as a foundational platform for complex manipulation tasks. However, legacy software frameworks lack the modularity and real-time responsiveness required for heavy extraterrestrial construction missions, such as assembling lunar or Martian habitats.

This project introduces an **Enhanced Robot Operating System (ROS/ROS 2) framework** tailored specifically for humanoid robots like NASA Robonaut 2 to execute precise, force-controlled construction tasks. By upgrading kinematic modeling, integrating whole-body control (WBC), enhancing real-time perception, and utilizing safety-certified middleware, the enhanced ROS framework enables R2 to handle heavy modular components, anchor structural elements, and construct resilient space habitats alongside or in place of human astronauts.

2. PROBLEM STATEMENT

Constructing habitats on extraterrestrial surfaces (e.g., Moon or Mars) involves unstructured terrain, microgravity or partial gravity constraints, and heavy lifting operations that exceed standard human endurance.

- **Limitations of Legacy Control Systems:** Traditional control architectures lack real-time synchronization between dual-arm manipulation, multi-contact balancing, and force/torque feedback.
- **Middleware Inefficiencies:** Standard ROS implementations face latency bottlenecks, lack deterministic real-time execution guarantees, and fail to handle abrupt sensor failures or payload shifts during heavy construction.
- **Lack of Integrated Spatial Planning:** Existing systems struggle with closed-loop tactile feedback required for interlocking habitat modules, handling raw regolith-based building materials, and operating power tools safely in extreme thermal and vacuum environments.

3. OBJECTIVES AND SCOPE

3.1 Objectives

- Upgrade the ROS middleware stack with real-time extensions (ROS 2 / DDS) for low-latency communication across R2's 42 Degrees of Freedom (DoF).
- Implement Whole-Body Control (WBC) algorithms integrating dual-arm dexterous manipulation with mobile base/leg stabilization for heavy payload transport.
- Integrate multi-modal perception pipelines (stereo vision, LiDAR, and custom six-axis load cells in the hands) to map habitat construction sites.
- Establish automated error recovery and safety-monitoring layers to prevent structural collapses or hardware overloads during construction tasks.

3.2 Scope

The design focuses on software simulation and hardware-in-the-loop verification using Robonaut 2's kinematics model. It covers ROS 2 node architecture, custom action servers for tool manipulation, force-torque control loops, and decentralized telemetry storage. Multi-robot collaborative habitat assembly is bounded within simulation environments (Gazebo/Ignition), while hardware execution targets R2's upper torso and dexterous end-effectors.

4. EXISTING AND PROPOSED SYSTEM

4.1 Existing Approach

- Relies on partitioned, mission-specific control software with limited code reusability across different robotic platforms.
- Teleoperation-heavy workflows introduce high communication latency between Earth and space assets, delaying critical habitat assembly decisions.
- Open-loop execution for heavy lifting leads to mechanical binding, joint saturation, and slippage when handling unstructured building blocks.

4.2 Proposed Approach

The proposed Enhanced ROS framework introduces real-time deterministic communication, closed-loop impedance control, and autonomous task execution specifically optimized for humanoid habitat construction.

Feature	Proposed Mechanism	Purpose
Real-Time Communication	ROS 2 DDS (Data Distribution Service)	Eliminate communication jitter and ensure deterministic command execution
Manipulation & Control	Affordance Templates & Whole-Body Control	Coordinate dual arms, torso, and end-effectors for heavy lifting
Tactile & Force Feedback	Custom 6-Axis Load Cell & Impedance Loops	Prevent structural damage during precise module interlocking
Perception Pipeline	Sensor Fusion (Stereo + Depth + LiDAR)	Real-time environment mapping and obstacle avoidance
Safety & Fault Tolerance	Watchdog nodes & automated state pruning	Isolate software/hardware faults instantly

5. SYSTEM REQUIREMENTS

Software and environment requirements to run the Enhanced ROS architecture for Robonaut 2:

Component	Technology / Environment	Role
Host Operating System	Ubuntu 22.04 LTS (ROS 2 Humble / Iron)	Primary robotics development and execution environment
Real-Time Kernel	Linux PREEMPT_RT patch	Ensures hard real-time execution for servo loops
Simulation Environment	Gazebo Fortress / Ignition	Physics-based lunar/Martian habitat simulation
Control Framework	MoveIt 2, ros2_control, OMPL	Motion planning and trajectory execution
Database & Telemetry	PostgreSQL & TimescaleDB	Storing teleoperation logs, sensor telemetry, and trial metrics
Visualization Interface	Foxglove Studio / RViz2	Real-time telemetry, point cloud, and state visualization

6. SYSTEM ARCHITECTURE

The system architecture follows a decentralized, modular node-based paradigm separated into four primary layers:

- **1. Perception Layer:** Aggregates data from Robonaut 2's visor cameras, infrared depth sensors, and joint encoders.
- **2. Planning & Decision Layer:** Handles task allocation, path generation via MoveIt 2, and affordance template execution.
- **3. Control & Execution Layer:** Executes Whole-Body Control (WBC) and impedance-based torque commands across R2's 54 servomotors.
- **4. Persistence & Presentation Layer:** Streams telemetry to PostgreSQL and visualizes operational metrics via Foxglove Studio/RViz2.

7. ALGORITHMS AND CORE CONCEPTS

7.1 Whole-Body Impedance Control

To handle heavy structural panels without damaging connectors, the framework uses operational space impedance control:

> **$\Lambda(x)\ddot{x} + \mu(x, \dot{x})\dot{x} + g(x) = F_{ext} - F_{control}$**

This formulation allows the robot's arms and torso to yield compliantly when contacting rigid habitat surfaces, preventing mechanical binding and jamming.

7.2 Affordance Templates for Tool and Module Handling

Building upon NASA's task-priming frameworks, affordance templates encapsulate the geometry and relative constraints required to manipulate standard construction tools, bolts, and interlocking habitat trusses.

7.3 Real-Time Task Prioritization

Using hierarchical null-space projection, higher-priority tasks (e.g., balance and collision avoidance) are solved first, while secondary tasks (e.g., tool orientation) utilize remaining degrees of freedom without compromising safety.

8. DATABASE DESIGN

The backend database schema logs execution metrics, system telemetry, and failure events during habitat construction trials.

Table	Important Fields	Purpose
Habitat Runs	run_id (PK), site_name, habitat_module_type, start_time, end_time, status	Tracks overall construction sessions
Task Trials	trial_id (PK), run_id (FK), subtask_name, execution_time_sec, success_flag	Tracks individual manipulation and assembly trials
Telemetry Logs	id (PK), trial_id (FK), timestamp, joint_torque_nm, cpu_temp_c, power_draw_w	Hardware health and energy consumption tracking

9. COMMUNICATION AND DATA FLOW

1. Mission objectives and habitat blueprint parameters are loaded into the ROS 2 action server.

2. The perception node streams point-cloud data to identify construction targets and obstacles.

3. The motion planner calculates collision-free trajectories for Robonaut 2's dual arms.

4. Joint velocity and torque commands are streamed via DDS topics to R2's motor controllers.

5. Force sensors provide high-frequency feedback to modify impedance trajectories in real time.

6. Telemetry and performance metrics are asynchronously logged to PostgreSQL.

10. IMPLEMENTATION PLAN

- **Phase 1:** Setup Ubuntu PREEMPT_RT kernel, ROS 2 workspace, and Gazebo habitat simulation environment.
- **Phase 2:** Import Robonaut 2 URDF/SRDF kinematic models into MoveIt 2 and configure ros2_control interfaces.
- **Phase 3:** Implement Whole-Body Control and dual-arm manipulation nodes.
- **Phase 4:** Integrate sensor fusion pipelines for stereo vision and hand-load cells.
- **Phase 5:** Build automated habitat module interlocking sequences using affordance templates.
- **Phase 6:** Deploy PostgreSQL telemetry logging and Foxglove visualization dashboards.
- **Phase 7:** Conduct end-to-end simulation testing and performance evaluation.

11. TESTING AND EVALUATION PLAN

- **Kinematic Accuracy Tests:** Measure end-effector positioning error under simulated lunar/Martian gravity profiles.
- **Payload Handling Benchmarks:** Test lifting and stabilizing modular habitat tiles weighing up to Robonaut 2's payload threshold (9 kg per arm).
- **Impedance Response Evaluation:** Introduce external disturbances during peg-in-hole insertion tasks to verify compliance and fault recovery.
- **Communication Latency Stress Tests:** Evaluate ROS 2 DDS packet drop rates and node recovery times under simulated deep-space communication delays.

12. EXPECTED OUTCOMES

- Fully verified Enhanced ROS 2 framework capable of controlling humanoid robots for intricate construction tasks.
- Demonstration of stable, force-compliant dual-arm manipulation in simulated habitat assembly scenarios.
- Comprehensive telemetry database and visualization suite for performance analysis and remote monitoring.
- Significant reduction in task failure rates through active impedance control and automated fault handling.

13. LIMITATIONS AND FUTURE ENHANCEMENTS

13.1 Limitations

- Simulation models approximate real-world regolith interaction and abrasive dust dynamics.
- Hardware testing is constrained by the availability of physical Robonaut 2 flight-spare hardware units.
- Bandwidth constraints in deep-space communication require high levels of onboard autonomy, limiting real-time Earth-based supervisor intervention.

13.2 Future Enhancements

- Integrate generative AI models for adaptive on-site path replanning when facing unforeseen structural blockages.
- Expand multi-robot coordination frameworks for collaborative habitat construction using multiple humanoids (e.g., Robonaut 2 and Valkyrie).
- Implement advanced tactile skin sensors across the upper torso for full-body collision avoidance.

14. CONCLUSION

The **Enhanced ROS for Humanoid Robots (NASA Robonaut 2) in Constructing Habitats** project provides a robust, real-time software foundation for space robotics. By combining ROS 2 architecture, Whole-Body Impedance Control, and tactile feedback mechanisms, this system bridges the gap between general-purpose humanoid dexterity and the rigorous demands of extraterrestrial infrastructure development. The architecture ensures safe, precise, and verifiable execution of complex assembly tasks, paving the way for sustainable human-robot collaborative space missions.